

Moments

Moments About Mean (central moment)

Central Moments are calculated using deviations from the **arithmetic mean** ($\bar{x}$). They are "central" because the mean is the center of gravity of the distribution. They are the most useful moments for describing the shape of a distribution because they are not affected by changes in the data's location (i.e., adding a constant to all values).

The r^{th} central moment is denoted by the Greek letter μ_r (mu with subscript r).

Formula for Ungrouped Data:

$$\mu_r = \frac{\sum (x_i - \bar{x})^r}{N}$$

Formula for Grouped Data:

$$\mu_r = \frac{\sum f_i (x_i - \bar{x})^r}{N}$$

Where:

- x_i = Individual observation (or midpoint of a class)
- $\bar{x}$ = The arithmetic mean
- f_i = Frequency of the observation
- N = Total frequency ($\sum f_i$)
- r = The order of the moment (1, 2, 3, 4, ...)

Interpretation of the First Four Central Moments:

- **First Central Moment (μ_1):**

$$\mu_1 = \frac{\sum f_i (x_i - \bar{x})^1}{N} = 0$$

The first central moment is **always zero**. This is a fundamental property of the arithmetic mean, as the sum of all deviations from the mean is always zero.

- **Second Central Moment (μ_2):**

$$\mu_2 = \frac{\sum f_i(x_i - \bar{x})^2}{N}$$

This is the **Variance** (σ^2). It is the most common measure of the data's dispersion or spread.

- **Third Central Moment** (μ_3):

$$\mu_3 = \frac{\sum f_i(x_i - \bar{x})^3}{N}$$

This moment is used to measure **Skewness** (asymmetry).

- If $\mu_3 > 0$, the distribution is **positively skewed** (long tail to the right).
- If $\mu_3 < 0$, the distribution is **negatively skewed** (long tail to the left).
- If $\mu_3 = 0$, the distribution is **symmetrical**.

- **Fourth Central Moment** (μ_4):

$$\mu_4 = \frac{\sum f_i(x_i - \bar{x})^4}{N}$$

This moment is used to measure **Kurtosis** (peakedness). It describes how "tailed" and "peaked" the distribution is compared to a normal (bell-shaped) distribution.



Moments About Any Arbitrary Number (Raw moment)

Raw Moments (also called non-central moments) are calculated using deviations from an **arbitrary point 'A'**. This point 'A' can be any number, but it is often chosen to simplify calculations (this is the basis of the "shortcut method" for calculating variance).

The r^{th} raw moment about point 'A' is denoted by μ_r (mu-prime).

Formula for Ungrouped Data:

$$\mu_r = \frac{\sum (x_i - a)^r}{N}$$

Formula for Grouped Data:

$$\mu_r = \frac{\sum f_i(x_i - a)^r}{N}$$

While easier to compute, raw moments are less descriptive than central moments. Their main purpose is to be used as a stepping stone to find the central moments.

Relationship between Central and Raw Moments:

The central moments (μ_r) can be calculated from the raw moments (μ_r) using the following conversion formulas (these are known as **Sheppard's Corrections** in some contexts, though the formulas themselves are general conversions):

- $\mu_1 = 0$ (by definition)
- $\mu_2 = \mu_2 - (\mu_1)^2$
- $\mu_3 = \mu_3 - 3\mu_1\mu_2 + 2(\mu_1)^3$
- $\mu_4 = \mu_4 - 4\mu_1\mu_3 + 6(\mu_1)^2\mu_2 - 3(\mu_1)^4$

The most important of these is the formula for μ_2 :

$$\mu_2 = (\mu_2) - (\mu_1)^2$$



Moments About Origin

Moments about the Origin are the most common type of **raw moment**. They are simply a special case where the arbitrary point 'A' is set to **zero** ($= 0$).

Formula for Ungrouped Data:

$$\mu_r(\text{ao0}) = \frac{\sum (x_i - 0)^r}{N} = \frac{\sum x_{ri}}{N}$$

Formula for Grouped Data:

$$\mu_r(\text{ao0}) = \frac{\sum f_i(x_i - 0)^r}{N} = \frac{\sum f_i x_{ri}}{N}$$

Interpretation of the First Two Raw Moments about the Origin:

- **First Raw Moment about Origin (μ_1):**

$$\mu_1 = \frac{\sum f_i x_{1i}}{N} = \frac{\sum f_i x_i}{N}$$

This is the formula for the **Arithmetic Mean** ($\bar{x}$). Therefore, the first moment about the origin is simply the mean of the distribution.

- **Second Raw Moment about Origin** (μ_2):

$$\mu_2 = \frac{\sum f_i x_{2i}}{N}$$

This is the "mean of the squares." It doesn't have a simple name on its own, but it is critical for calculating variance.

Putting It All Together (The "Shortcut Formula" for Variance):

Let's use the conversion formula we learned earlier: $\mu_2 = \mu_2 - (\mu_1)^2$

Now, let's use the moments about the origin ($= 0$):

- $\mu_2 = \text{Variance } (\sigma^2)$
- $\mu_1 = \text{First raw moment about origin} = \text{Mean } (\bar{x})$
- $\mu_2 = \text{Second raw moment about origin} = \frac{\sum f_i x_{2i}}{N}$

By substituting these in, we get the standard computational formula for variance:

$$\sigma^2 = \left(\frac{\sum f_i x_{2i}}{N} \right) - (\bar{x})^2$$

This proves that the "shortcut formula" for variance is just the formula for converting the second raw moment about the origin into the second central moment.



Moment

Meaning Of Moment

In statistics, a **moment** is a quantitative measure that describes the shape of a data distribution. The term is borrowed from physics, where "moment" refers to a force's tendency to cause rotation around a point (like the center of mass).

In statistics, we use moments to describe the shape of a distribution relative to a point.

- **Moments about the Origin (Raw Moments):** When the moments are calculated around the value '0'. The first raw moment is the Arithmetic Mean ($\bar{x}$).
- **Moments about the Mean (Central Moments):** When the moments are calculated around the Arithmetic Mean ($\bar{x}$). These are more common and are used to describe the *shape* of the distribution, independent of its location.

We will focus on **Central Moments**, denoted by the Greek letter μ (mu).

- **First Central Moment (μ_1):** Measures the center of gravity. It is always **zero**.
- **Second Central Moment (μ_2):** Measures the **Variance**. It describes the *spread* or *width* of the distribution.
- **Third Central Moment (μ_3):** Measures **Skewness**. It describes the *asymmetry* of the distribution (whether it leans to the left or right).
- **Fourth Central Moment (μ_4):** Measures **Kurtosis**. It describes the *peakedness* and *tail thickness* of the distribution.

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Moments About Mean For Individual Series

These are the central moments calculated for ungrouped data (a simple list of values).

Formula

The r -th moment about the mean (μ_r) is defined as the mean of the r -th power of the deviations from the mean.

$$\mu_r = \frac{\sum (x - \bar{x})^r}{N}$$

- **First Moment (μ_1):**

$$\mu_1 = \frac{\sum (x - \bar{x})^1}{N}$$

- **Second Moment (μ_2):**

$$\mu_2 = \frac{\sum (x - \bar{x})^2}{N}$$

- **Third Moment (μ_3):**

$$\mu_3 = \frac{\sum (x - \bar{x})^3}{N}$$

Tricks (Key Properties)

1. The **First Moment about the mean (μ_1) is always zero**. This is because $\sum (x - \bar{x}) = 0$ (the sum of deviations from the mean is always zero), so $\frac{0}{N} = 0$.
2. The **Second Moment about the mean (μ_2) is the Variance (σ^2)**. This is the definition of variance.

Relation of Third Moment About Mean And Variance And Standard Deviation

- **Variance (σ^2):** Is exactly equal to the Second Moment about the mean (μ_2).

$$\implies \sigma^2 = \mu_2$$

- **Standard Deviation (σ):** Is the square root of the Variance, so it is the square root of the Second Moment about the mean.

$$\implies \sigma = \sqrt{\mu_2}$$

- **Third Moment (μ_3):** This value is used to calculate **skewness**. It is not directly related to variance by a simple formula, but they are both moments of the same distribution.
 - If $\mu_3 = 0$, the distribution is symmetrical.
 - If $\mu_3 > 0$ (positive), the distribution is positively (right) skewed.
 - If $\mu_3 < 0$ (negative), the distribution is negatively (left) skewed.

Solved Example

Question: Find the first three moments about the mean for the following data: 2, 3, 4, 5, 6.

Solution:

Here $N = 5$.

Step 1: Calculate the Mean ($\bar{x}$).

$$\bar{x} = \frac{\sum x}{N} = \frac{2 + 3 + 4 + 5 + 6}{5} = \frac{20}{5} = 4$$

Step 2: Create a calculation table.

We need the sums of the deviations, the squared deviations, and the cubed deviations.

Let $d = (x - \bar{x}) = (x - 4)$.

	$d = (x - 4)$	$d^2 = (x - 4)^2$	$d^3 = (x - 4)^3$
2	-2	4	-8
3	-1	1	-1
4	0	0	0
5	1	1	1
6	2	4	8
Total	$\sum d = 0$	$\sum d^2 = 10$	$\sum d^3 = 0$

Step 3: Calculate the moments.

- **First Moment (μ_1):**

$$\mu_1 = \frac{\sum d}{N} = \frac{0}{5} = 0$$

- **Second Moment (μ_2):**

$$\mu_2 = \frac{\sum d^2}{N} = \frac{10}{5} = 2$$

- **Third Moment (μ_3):**

$$\mu_3 = \frac{\sum d^3}{N} = \frac{0}{5} = 0$$

Interpretation:

- $\mu_1 = 0$, which is always true.
- $\mu_2 = 2$. This means the **Variance** (σ^2) is 2.
- $\mu_3 = 0$. This indicates that the data is **perfectly symmetrical**, which we can see (4 is the center, 3 and 5 are balanced, 2 and 6 are balanced).



Moments About Mean For Discrete Or Continuous Distribution

For data with frequencies, the logic is the same, but we must multiply each deviation by its frequency (f).

Formula

The r -th moment about the mean (μ_r) for a frequency distribution is:

$$\mu_r = \frac{\sum f(x)^r}{N}$$

- For **Discrete Series**: $x = (x_i - \bar{x})$
- For **Continuous Series**: $x = (x_i - \bar{x})$, where $\bar{x}$ is the mid-point.

Formulas:

- **First Moment** (μ_1):

$$\mu_1 = \frac{\sum f(x)}{N}$$

- **Second Moment** (μ_2):

$$\mu_2 = \frac{\sum f(x)^2}{N}$$

- **Third Moment** (μ_3):

$$\mu_3 = \frac{\sum f(x - \bar{x})^3}{N} \text{ or } \frac{\sum f(x - \bar{x})^3}{N}$$

Tricks (Key Properties)

These are identical to the individual series:

1. The **First Moment about the mean** (μ_1) is always zero.
2. The **Second Moment about the mean** (μ_2) is the **Variance** (σ^2).

Relation of Third Moment About Mean And Variance And Standard Deviation

This relationship is also identical to the individual series:

- **Variance** (σ^2) = μ_2
- **Standard Deviation** (σ) = $\sqrt{\mu_2}$
- **Third Moment** (μ_3) is the primary measure used to calculate skewness.

Solved Example (Discrete Series)

Question: Find the first three moments about the mean for the following distribution.

	1	2	3	4	5
f	2	4	6	4	2

Solution:

Step 1: Calculate N and the Mean ($\bar{x}$).

We first need $\sum f$ and $\sum fx$.

	f	fx
1	2	2
2	4	8

	f	f
3	6	18
4	4	16
5	2	10
Total	$N = 18$	$\sum f = 54$

- **Mean ($\bar{x}$):**

$$\bar{x} = \frac{\sum f}{N} = \frac{54}{18} = 3$$

Step 2: Create the main calculation table.

Let $u = (x - \bar{x}) = (x - 3)$.

	f	$u = (x - 3)$	u^2	u^3	$f \cdot u$	$f \cdot u^2$	$f \cdot u^3$
1	2	-2	4	-8	$2 \times -2 = -4$	$2 \times 4 = 8$	$2 \times -8 = -16$
2	4	-1	1	-1	$4 \times -1 = -4$	$4 \times 1 = 4$	$4 \times -1 = -4$
3	6	0	0	0	$6 \times 0 = 0$	$6 \times 0 = 0$	$6 \times 0 = 0$
4	4	1	1	1	$4 \times 1 = 4$	$4 \times 1 = 4$	$4 \times 1 = 4$
5	2	2	4	8	$2 \times 2 = 4$	$2 \times 4 = 8$	$2 \times 8 = 16$
Total	$N = 18$				$\sum f \cdot u = 0$	$\sum f \cdot u^2 = 24$	$\sum f \cdot u^3 = 0$

Step 3: Calculate the moments.

- **First Moment (μ_1):**

$$\mu_1 = \frac{\sum f \cdot u}{N} = \frac{0}{18} = 0$$

- **Second Moment (μ_2):**

$$\mu_2 = \frac{\sum f \cdot u^2}{N} = \frac{24}{18} = 1.333$$

- **Third Moment (μ_3):**

$$\mu_3 = \frac{\sum f \cdot u^3}{N} = \frac{0}{18} = 0$$

Interpretation:

- $\mu_1 = 0$, as expected.
- $\mu_2 = 1.333$. This means the **Variance** (σ^2) is **1.333**.
- $\mu_3 = 0$. This indicates the distribution is **symmetrical**.

Here are the notes on Moments about the Origin and their relation to Moments about the Mean.

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Moments about Origin in Statistics

Moments about the Origin, also known as **Raw Moments**, are the moments calculated around the value '0'. They are denoted by a prime symbol (μ_r).

These moments describe the distribution in its "raw" form, without centering it first. The first moment about the origin is simply the arithmetic mean.

Formula

The r -th moment about the origin (μ_r) is defined as the mean of the r -th power of the values.

- **For Individual Series:**

$$\mu_r = \frac{\sum x^r}{N}$$

- **For Discrete/Continuous Series:**

$$\mu_r = \frac{\sum f^r}{N} \text{ or } \frac{\sum f^r}{N} (\text{if } N = \sum f)$$

The first four raw moments are:

1. **First Moment** (μ_1):

$$\mu_1 = \frac{\sum}{N} = \text{-(eieiMean)}$$

2. **Second Moment (μ_2):**

$$\mu_2 = \frac{\sum^2}{N}$$

3. **Third Moment (μ_3):**

$$\mu_3 = \frac{\sum^3}{N}$$

4. **Fourth Moment (μ_4):**

$$\mu_4 = \frac{\sum^4}{N}$$

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Solved Example

Question: Find the first three moments about the origin for the following data: 2, 3, 4, 5, 6. (This is the same data from our previous example on central moments).

Solution:

Here $N = 5$. We need to find the sum of , ², and ³.

Calculation Table:

	²	³
2	4	8
3	9	27
4	16	64
5	25	125
6	36	216
$\sum = 20$	$\sum^2 = 90$	$\sum^3 = 440$

Calculation:

- **First Moment (μ_1):**

$$\mu_1 = \frac{\sum}{N} = \frac{20}{5} = 4$$

(This is the Arithmetic Mean, $\bar{x}$)

- **Second Moment (μ_2):**

$$\mu_2 = \frac{\sum^2}{N} = \frac{90}{5} = 18$$

- **Third Moment (μ_3):**

$$\mu_3 = \frac{\sum^3}{N} = \frac{440}{5} = 88$$

Answer: The first three moments about the origin are $\mu_1 = 4$, $\mu_2 = 18$, and $\mu_3 = 88$.

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Relation between Moment about Mean and Moment about Origin

Moments about the Mean (Central Moments, μ_r) can be expressed in terms of **Moments about the Origin** (Raw Moments, μ_r).

This is extremely useful because it is often easier to calculate the raw moments first (summing x , x^2 , x^3 , etc.) and then use these conversion formulas to find the central moments, rather than calculating all the deviations ($x - \bar{x}$).

Formula

Remember that $\mu_1 = \bar{x}$

- **First Central Moment:**

$$\mu_1 = 0 \text{ (definition)}$$

- **Second Central Moment (Variance):**

$$\mu_2 = \mu_2 - (\mu_1)^2$$

(This is the most important one. It means: Variance = (Mean of the squares) - (Square of the mean))

- **Third Central Moment:**

$$\mu_3 = \mu_3 - 3(\mu_2)(\mu_1) + 2(\mu_1)^3$$

- **Fourth Central Moment:**

$$\mu_4 = \mu_4 - 4(\mu_3)(\mu_1) + 6(\mu_2)(\mu_1)^2 - 3(\mu_1)^4$$

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Solved Example

Question: Using the data from the previous two examples, confirm the relationship between the moments.

Data: 2, 3, 4, 5, 6

From our previous calculations, we know:

- **Moments about Origin:** $\mu_1 = 4$, $\mu_2 = 18$, $\mu_3 = 88$
- **Moments about Mean:** $\mu_1 = 0$, $\mu_2 = 2$, $\mu_3 = 0$

Let's use the conversion formulas to see if we get the correct moments about the mean.

1. Calculate μ_2 (Variance):

- **Formula:** $\mu_2 = \mu_2 - (\mu_1)^2$
- **Calculation:**

$$\mu_2 = 18 - (4)^2$$

$$\mu_2 = 18 - 16$$

$$\mu_2 = 2$$

- **Check:** This matches our direct calculation of μ_2 in the previous lesson.

2. Calculate μ_3 :

- **Formula:** $\mu_3 = \mu_3 - 3(\mu_2)(\mu_1) + 2(\mu_1)^3$
- **Calculation:**

$$\mu_3 = 88 - 3(18)(4) + 2(4)^3$$

$$\mu_3 = 88 - 3(72) + 2(64)$$

$$\mu_3 = 88 - 216 + 128$$

$$\mu_3 = 216 - 216$$

$$\mu_3 = 0$$

- **Check:** This also matches our direct calculation of μ_3 in the previous lesson.

This example proves that the conversion formulas work and are a valid alternative method for finding the central moments.

Here are the notes on Karl Pearson's coefficients and practice questions on moments.



Moment About Mean Questions

Karl Pearson Coefficients

Karl Pearson's coefficients are dimensionless measures used to describe the **shape** of a distribution. They use the central moments (μ_r) to quantify skewness and kurtosis.

Beta Coefficients

1. **Beta 1 (β_1):** This coefficient measures **skewness** (the asymmetry of the distribution). It is based on the square of the third moment, so its value is always non-negative.

- **Formula:**

$$\gamma_1 = \frac{\mu_{23}}{\mu_{32}}$$

- **Interpretation:**

- If $\gamma_1 = 0$, the distribution is symmetrical.
- If $\gamma_1 \neq 0$, the distribution is skewed (asymmetrical). The larger the value, the greater the skewness.

2. **Beta 2 (β_2):** This coefficient measures **kurtosis** (the "peakedness" or "tail-heaviness" of the distribution). It compares the distribution's shape to a normal distribution.

- **Formula:**

$$\beta_2 = \frac{\mu_4}{\mu_{22}}$$

- **Interpretation:**

- If $\beta_2 = 3$, the distribution has the same kurtosis as a normal distribution (mesokurtic).
- If $\beta_2 > 3$, the distribution is more peaked with heavier tails than a normal distribution (leptokurtic).
- If $\beta_2 < 3$, the distribution is flatter with lighter tails than a normal distribution (platykurtic).

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Gamma Coefficients

Gamma coefficients are related to Beta coefficients but are often preferred for interpretation.

1. **Gamma 1 (γ_1):** This is the **coefficient of skewness**. It is the square root of β_1 and retains the original sign of μ_3 . This makes it more descriptive than β_1 .

- **Formula:**

$$\gamma_1 = \sqrt{\beta_1} = \frac{\mu_3}{\mu_{3/22}} = \frac{\mu_3}{\sigma^3}$$

- **Interpretation:**

- If $\mu_1 = 0$, the distribution is symmetrical.
- If $\mu_1 > 0$, it is positively (right) skewed.
- If $\mu_1 < 0$, it is negatively (left) skewed.

2. **Gamma 2 (μ_2):** This is the **coefficient of excess kurtosis**. It measures kurtosis relative to the normal distribution (which has $\mu_2 = 3$).

- **Formula:**

$$\mu_2 = \mu_2 - 3$$

- **Interpretation:**

- If $\mu_2 = 0$, the distribution is mesokurtic.
- If $\mu_2 < 0$, the distribution is leptokurtic.
- If $\mu_2 > 0$, the distribution is platykurtic.



Questions Practice

Question 1

The First Four Moments Of A Distribution About The Value '4' of the variable are -1.5, 17, -30 and 108. Find The Moments About Mean, About Origin , Beta 1 and Beta 2. Also Find The Moments About The Point $x = 2$.

Solution:

First, let's identify what is given. These are moments about an arbitrary point (or provisional moments), $a = 4$.

Let's call these provisional moments $\mu_r(a)$.

- $a = 4$
- $\mu_1(a) = -1.5$
- $\mu_2(a) = 17$
- $\mu_3(a) = -30$

- $\mu_4() = 108$

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1. Moments About Mean (Central Moments, μ_r)

We use the conversion formulas that relate moments about an arbitrary point to the central moments (μ_r) about the mean.

- μ_1 :
 $\mu_1 = 0$ (By definition, the first moment about the mean is always zero).
- μ_2 (Variance):

$$\mu_2 = \mu_2() - (\mu_1())^2$$

$$\mu_2 = 17 - (-1.5)^2$$

$$\mu_2 = 17 - 2.25$$

$$\mu_2 = 14.75$$

- μ_3 :

$$\mu_3 = \mu_3() - 3(\mu_2())(\mu_1()) + 2(\mu_1())^3$$

$$\mu_3 = (-30) - 3(17)(-1.5) + 2(-1.5)^3$$

$$\mu_3 = -30 - 3(-25.5) + 2(-3.375)$$

$$\mu_3 = -30 + 76.5 - 6.75$$

$$\mu_3 = 39.75$$

- μ_4 :

$$\mu_4 = \mu_4() - 4(\mu_3())(\mu_1()) + 6(\mu_2())(\mu_1())^2 - 3(\mu_1())^4$$

$$\mu_4 = 108 - 4(-30)(-1.5) + 6(17)(-1.5)^2 - 3(-1.5)^4$$

$$\mu_4 = 108 - 4(45) + 6(17)(2.25) - 3(5.0625)$$

$$\mu_4 = 108 - 180 + 229.5 - 15.1875$$

$$\mu_4 = 142.3125$$

Answer (Part 1): The moments about the mean are $\mu_1 = 0$, $\mu_2 = 14.75$, $\mu_3 = 39.75$, and $\mu_4 = 142.3125$.

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2. Moments About Origin (Raw Moments, μ_r)

To find the moments about the origin (0), we first need the true mean (μ).

$$\mu = \sum x_i f_i / \sum f_i = 4 + (-1.5) = 2.5$$

The first moment about the origin (μ_1) is the mean itself:

$$\mu_1 = \mu = 2.5$$

Now we use the formulas to convert *from* central moments (μ_r) *to* raw moments (μ_r). Here

$$\mu_1 = \mu = 2.5.$$

- μ_2 (Raw):

$$\mu_2 = \mu_2 + (\mu_1)^2$$

$$\mu_2 = 14.75 + (2.5)^2$$

$$\mu_2 = 14.75 + 6.25$$

$$\mu_2 = 21$$

- μ_3 (Raw):

$$\mu_3 = \mu_3 + 3(\mu_2)(\mu_1) + (\mu_1)^3$$

$$\mu_3 = 39.75 + 3(14.75)(2.5) + (2.5)^3$$

$$\mu_3 = 39.75 + 110.625 + 15.625$$

$$\mu_3 = 166$$

- μ_4 (Raw):

$$\mu_4 = \mu_4 + 4(\mu_3)(\mu_1) + 6(\mu_2)(\mu_1)^2 + (\mu_1)^4$$

$$\mu_4 = 142.3125 + 4(39.75)(2.5) + 6(14.75)(2.5)^2 + (2.5)^4$$

$$\mu_4 = 142.3125 + 397.5 + 6(14.75)(6.25) + 39.0625$$

$$\mu_4 = 142.3125 + 397.5 + 553.125 + 39.0625$$

$$\mu_4 = 1132$$

Answer (Part 2): The moments about the origin are $\mu_1 = 2.5$, $\mu_2 = 21$, $\mu_3 = 166$, and

$$\mu_4 = 1132.$$

3. Beta 1 (β_1) and Beta 2 (β_2)

We use the central moments (μ_r) that we found in the first section.

- $\mu_2 = 14.75$
- $\mu_3 = 39.75$
- $\mu_4 = 142.3125$
- **Beta 1 (β_1):**

$$\beta_1 = \frac{\mu_3}{\mu_2}$$

$$\beta_1 = \frac{(39.75)^2}{(14.75)^3}$$

$$\beta_1 = \frac{1580.0625}{3211.546875}$$

$$\beta_1 \approx 0.492$$

- **Beta 2 (β_2):**

$$\beta_2 = \frac{\mu_4}{\mu_2^2}$$

$$\beta_2 = \frac{142.3125}{(14.75)^2}$$

$$\beta_2 = \frac{142.3125}{217.5625}$$

$$\beta_2 \approx 0.654$$

Answer (Part 3): $\beta_1 \approx 0.492$ (skewed) and $\beta_2 \approx 0.654$ (very platykurtic, or flat).

4. Moments About The Point $x = 2$

This is a "change of origin" problem. We have moments about $x = 4$ and want to find moments about $x = 2$.

Let $h = - = 4 - 2 = 2$.

The relationship between the deviations is $= (-) = (-) + (-) = + h$.

So, $\mu_r() = \frac{\sum(+h)^r}{N}$. This gives us the following conversion formulas:

- $\mu_1(= 2)$:

$$\mu_1() = \mu_1() + h$$

$$\mu_1() = -1.5 + 2$$

$$\mu_1() = 0.5$$

- $\mu_2(= 2)$:

$$\mu_2() = \mu_2() + 2h\mu_1() + h^2$$

$$\mu_2() = 17 + 2(2)(-1.5) + (2)^2$$

$$\mu_2() = 17 - 6 + 4$$

$$\mu_2() = 15$$

- $\mu_3(= 2)$:

$$\mu_3() = \mu_3() + 3h\mu_2() + 3h^2\mu_1() + h^3$$

$$\mu_3() = -30 + 3(2)(17) + 3(2)^2(-1.5) + (2)^3$$

$$\mu_3() = -30 + 102 + 3(4)(-1.5) + 8$$

$$\mu_3() = -30 + 102 - 18 + 8$$

$$\mu_3() = 62$$

- $\mu_4(= 2)$:

$$\mu_4() = \mu_4() + 4h\mu_3() + 6h^2\mu_2() + 4h^3\mu_1() + h^4$$

$$\mu_4() = 108 + 4(2)(-30) + 6(2)^2(17) + 4(2)^3(-1.5) + (2)^4$$

$$\mu_4() = 108 - 240 + 6(4)(17) + 4(8)(-1.5) + 16$$

$$\mu_4() = 108 - 240 + 408 - 48 + 16$$

$$\mu_4() = 244$$

Answer (Part 4): The moments about the point $x = 2$ are 0.5, 15, 62, and 244.

Question 2

The First Three Moments Of A Distribution About The Value '2' of the variable are 1, 16 and -40. Show That The Mean Is 3, Variance is 15 and $\mu_3 = -86$.

Solution:

First, let's identify the given information. These are provisional moments about $= 2$.

- $= 2$
- $\mu_1() = 1$
- $\mu_2() = 16$
- $\mu_3() = -40$

1. Show Mean is 3

The formula for the Arithmetic Mean ($\bar{x}$) based on a provisional mean (a) is:

$$\bar{x} = a + \mu_1()$$

- **Calculation:**

$$\bar{x} = 2 + 1$$

$$\bar{x} = 3$$

Result: The Mean is 3. (Proven)

2. Show Variance is 15

The Variance (σ^2) is the second central moment (μ_2). We use the formula to convert from provisional moments to the second central moment.

$$\mu_2 = \mu_2() - (\mu_1())^2$$

- **Calculation:**

$$\mu_2 = 16 - (1)^2$$

$$\mu_2 = 16 - 1$$

$$\mu_2 = 15$$

Result: The Variance is 15. (Proven)

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3. Show $\mu_3 = -86$

μ_3 is the third central moment. We use the conversion formula.

$$\mu_3 = \mu_3() - 3(\mu_2())(\mu_1()) + 2(\mu_1())^3$$

- **Calculation:**

$$\mu_3 = (-40) - 3(16)(1) + 2(1)^3$$

$$\mu_3 = -40 - 48 + 2(1)$$

$$\mu_3 = -40 - 48 + 2$$

$$\mu_3 = -88 + 2$$

$$\mu_3 = -86$$

Result: $\mu_3 = -86$. (Proven)